Low-Cost, Multisensor Nondestructive Banana Ripeness Estimation Using Machine Learning

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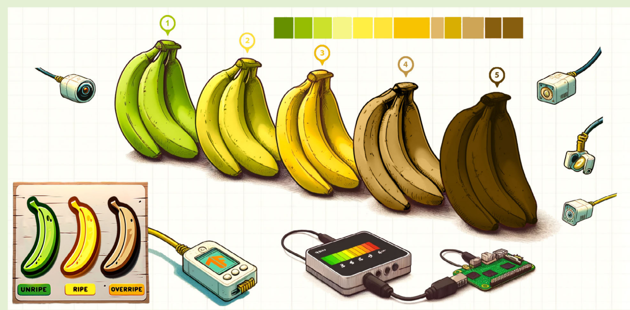
***Abstract*—Ripeness is a key metric in the growth, distribution, and sale of fruits. Present methods to determine a given fruit's ripeness rely on slow, labor intensive, and destructive methods that only provide an indication of crop-wide ripeness. A digital, low-cost, and nondestructive method would provide instant access to accurate ripeness data whenever required. In this article, a methodology is proposed for estimation of banana ripeness using an array of low-cost sensing modalities including cameras, spectrometers, volatile organic compound (VOC), and environmental sensors. Data were collected over five periods, each 35 days long (on average), using a setup designed for this systematic data collection. The banana's ripeness is classified into one of five stages using a range of machine-learning (ML) algorithms such as support vector machines (SVMs), random forest (RF), *K*-nearest neighbors (KNN), gradient boosting classifiers (GBCs), and artificial neural networks (ANNs), which were trained and tested against datasets constructed from different combinations of sensor data. RF and GBCs were found to be the most accurate at 99.95% each when using data from all available sensors arrays.**



***Index Terms*—Classification, computational intelligence, fruit ripeness, multimodal sensing.**

I. Introduction

BANANAS are the world's favorite fruit, with over 100 billion consumed annually, making up 75% of the annual global tropical fruit trade. The U.K. alone consumes 5 billion bananas per year, while Brazil, China, India, and Indonesia consume 50 billion [1], [2]. Identification and control of banana ripeness is therefore a key aspect of their production, distribution and sale, helping ensure they are harvested at the optimum maturity to ensure damage-free transport and perfect in-store ripeness. Bananas are one of the U.K.'s top-5 most wasted household foods (190 000 tons annually), and reducing supply-chain wastage is essential for reduction of global food wastage [3].

Current methods of fruit ripeness estimation involve measurement of the fruit's sugar and dry matter content by cutting open the fruit and extracting its juices [4]. This process must not only discard the tested fruit, but also only provides an *approximation* of crop-wide fruit maturity, allowing immature fruit to be harvested early, resulting in more wasted produce.

The physiological, biochemical, and organoleptic changes fruit undergo during the ripening process vary significantly between fruit types and varieties of the same fruit. Biochemical changes such as increased respiration, chlorophyll degradation, biosynthesis of carotenoids, anthocyanins, essential oils, and flavor and aroma components, increased activity of cell wall-degrading enzymes, and a transient increase in ethylene production occur universally during the ripening process [5], [6]. These changes can be detected and measured with nondestructive methods that can be used across an entire crop without generating wastage. Examples of nondestructive methods involve techniques such as Red Green Blue (RGB) imagery, hyperspectral imagery, or volatile organic compound (VOC) measurement in combination with machine learning (ML) algorithms to automatically and quickly classify the banana ripeness. The implementation of an automated system allows growers and supply-chain operatives to generate an almost instantaneous and accurate measurement of fruit maturity, enabling quick and accurate decision making at any time.

In this article, we present a methodology for multisensor laboratory data collection for the determination of banana ripeness, incorporating eight sensors of three types (optical, gaseous, and environmental) measuring five bananas over a

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total period of approximately 175 days using a specially designed test-rig. We present a data processing pipeline for the generation of appropriate datasets and demonstrate the use of several ML algorithms on these datasets for the purposes of classification of banana ripeness. The proposed multisensor approach and ML methods show accuracies that surpass those achieved in the literature. We also propose further developments that may help to reduce the aforementioned problems of global food wastage.

The remainder of this article is organized as follows. Section II describes the related work. Section III presents the methodologies for data collection, ripeness classification, and data analysis. Section IV shows the results and analysis of the proposed approach. Sections V and VI present the discussion and conclusion from this work, respectively.

II. Related Work

This section presents related works in sensors, ML methods and data processing algorithms used for the recognition of fruit ripeness. In the literature, fruit ripeness recognition has used a different number of ripeness stages, ranging from three to seven stages [7], [8], [9], [10]. In most cases, a rationale for the number of ripeness stages used was not provided, with the exception of the approaches in [11] and [12], both of which used a seven-stage framework. In previous works, the fruit ripeness recognition used common visual indicators, such as color [13], brown spot development, and Tamura statistical texture features [14], which are used as proxy for the fruit's biological ripeness.

A. Methods of Fruit Ripeness Detection

There are numerous nondestructive approaches available for detecting and quantifying fruit maturation (see Table I). Of the identified related works, the studies in [15] and [13] utilize RGB imaging, some form of spectroscopy is used in [7], [12], [13], [16], and [17], hyperspectral or multispectral imaging are used in [8], and tin-oxide-based VOC sensors are studied in [18]. The abundance of RGB imaging-based works is likely due to the availability of high-quality cameras and convolutional neural networks (CNNs). Such systems tend to achieve accuracies higher than 90%, but lack novelty as they only utilize one sensing modality.

Several novel nondestructive techniques were also identified, such as works using fruit skin characteristics (e.g., gloss, roughness, firmness, and starch/water content) [11], [12], laser-backscatter imaging, 5 GHz Wi-Fi signals, measurement of ultrasonic surface absorption [19], measurement of dielectric capacitance [20], and automated capacitive touch [21] all to classify the target fruit ripeness. These works focus on providing a proof-of-concept for the developed sensor, but they do not make use of ML techniques to process the collected data. Instead, they preferred to identify a relationship between measurement and obvious fruit ripeness changes. Consequently, they do not perform any accuracy testing of their systems, except for [9] and [21] which provide an accuracy of 90% and 88%, respectively, and [22] which uses an artificial neural network (ANN) to achieve a prediction accuracy of 95.5%. The most accurate methods presented in Table I utilize RGB imagery as their primary sensing method.

TABLE I
Related Fruit Ripeness and Classification Works

Ref.	Fruit Under Test	Sensing Method	Ripeness Stages Identified	ML Model's Deployed	Model Accuracy
[14]	Banana	RGB Imaging	4	ANN, SVM, Naïve Bayes, KNN, Decision Tree, Discriminant Analysis Classifiers	97.75% (ANN), 96.6% (SVM), 95.5% (Naïve Bayes), 95.5% (KNN), 97.75% (DT), 97.75% (DAC)
[24]	Banana	RGB Imaging	3	PCA, KNN, SVM, Decision Tree	96.6% (SVM & KNN), 95.5% (DT)
[25]	Banana	RGB Imaging	4	CNNs (MobileNet V2 & NASNetMobile)	96.18% (MobileNet V2), 90.84% (NASNetMobile)
[22]	Banana	Laser Backscatter Imaging	7	ANN	95.5%
[10]	Banana	Tin-Oxide VOC	7	PCA, Fuzzy ARTMAP, LVQ, MLP	92%
[26]	Banana	RGB Imaging	4	CNN & eXtreme Gradient Boosting, SVC, Gaussian Naïve Bayesian Classifier, KNN	91.25% (CNN & XgBoost), 78.75% (SVC), 79% (GNBC), 82.5% (KNN)
[27]	Banana	Hyperspectral & RGB Imaging	N/A	DNN, RF, MPL	90% (RGB only), 98.74% (HS only), 89.49% (HS only)
[29]	Strawberry	Hyperspectral Imaging	4	DNN	84%
[9]	Kiwi & Avocado	5GHz WiFi (> 600MHz)	4	N/A	90%

The works that measure VOCs utilize infrared (IR) or near infra-red (NIR) spectroscopy to analyze headspace gases, which are siphoned from an airtight container holding the target fruit into a gas cell for analysis [16], [23]. This methodology allows for complex chemometric analysis of the VOCs produced by the target fruit, and even enables fruit variety and ripeness/spoilage stage identification [17]. The works in [18] and [10] used specific VOC sensors (amongst others) directly measuring the target fruit. This prevents detailed chemometric analysis of the headspace gases, but does provide a more cost-effective and field-ready solution.

None of these works attempt to use multiple sensors of different types to classify the target fruit's ripeness, instead focusing on one sensor only or one sensor type. The work presented in this article makes use of a multimodal array of sensors, combining some of the approaches identified with ML to classify fruit ripeness with improved accuracy.

B. Data Processing in Fruit Ripeness Classification

The greatest use of ML was found to be amongst the works utilizing RGB imagery [13], [14], [15], [24], [25], [26], [27], [28]. Naturally, the use of CNNs featured frequently in this group, however, k-Nearest Neighbors(KNN) and support vector machines (SVMs) methods were also frequently used and reported good accuracies (78.75%–96.6%) [14], [15], [24], [25], [26]. Where works did not use RGB imagery as their primary sensor, ANNs or deep neural networks (DNNs) were generally employed, although both [10], [13] utilize versions of fuzzy logic machines. Table I shows that decision trees (DTs) and Naïve Bayes were also commonly used, typically achieving accuracies in the 95%–97.75% range [14], [24], [26], [27]. Most works making use of ML presented results from multiple methods, enabling a comparison of the most effective and accurate [10], [14], [24], [25], [26], [27].

(a) (b) (c) (d) (e) (f)

Fig. 1. Sensors used in this work. (a) BME280 temperature, pressure, and humidity sensor. (b) Pi Cam V2. (c) MapIR Survey3N OCN hyperspectral camera. (d) Figaro TGS2620/TGS2602 VOC sensor. (e) SGP40 VOC sensor. (f) Sparkfun TRIAD AS7265x spectrometer.

There is no obvious trend regarding the most accurate method. The majority of the identified works do not use ML to combine data from multiple sensor sources and as such cannot compare the accuracies achieved between different sets of sensor data. The work presented in this article conducts this comparison, as well as a comparison of the accuracy of all ML algorithms identified in the related literature for each sensor set, achieving superior accuracies than the related works.

C. Alternative Uses for Selected Sensors

The sensors selected for this work have demonstrated efficacy in a number of different applications. For example, the *Sparkfun TRIAD AS7265x* spectrometer has been employed in the following three works: 1) an IoT device to identify milk adulteration using ML [30]; 2) identification of weeds in agriculture [31]; and 3) creating a low-cost multispectral camera [32].

The *Figaro TGS2620* VOC sensor has been used in the following three unrelated works: 1) a wireless gas detection system for monitoring environmental pollution [33]; 2) a low-cost alcohol-hydration-meter [34]; and 3) a cardio-metabolic disease prevention system that measures patient breath [35]. Meanwhile, the *TGS2602* VOC sensor is typically used only in gas sensor applications. The *MapIR Survey3N OCN* hyperspectral camera has been used in: 1) mapping of geothermal features in volcanoes [36]; 2) mapping of groundwater level and soil moisture of a peat bog [37]; 3) informing generation of a vegetation index weighted canopy volume model for soybean biomass [38]; and 4) estimation of evapotranspiration [39].

Table I shows relevant works categorized by the fruit studied, sensors, ripeness stages, ML models, and accuracy.

III. Methodology

A. Sensor Selection

The optical, gaseous, and environmental sensors shown in Fig. 1 were utilized to maximize the variety of data collected throughout the fruit ripeness phases.

1) Optical Sensors: A Raspberry Pi RGB camera [see Fig. 1(b)] was used to provide a visual indicator of the fruit's general state. A *MapIR Survey3N OCN* hyperspectral camera [see Fig. 1(c)] (490, 615, and 808 nm) was used to provide additional optical data beyond the visible region. A *Sparkfun TRIAD AS7265x* spectrometer [see Fig. 1(f)] (detection range: 410–940 nm) was used to detect the fruit skin's spectral response.

2) Gaseous Sensors: The following sensors were used to measure gaseous outputs from the fruit; the *Figaro TGS2602* (to monitor air quality, VOCs and odors) [see Fig. 1(d)], the *TGS2620* (to detect organic solvent vapors and alcohols) [see Fig. 1(d)] and the *Sensiron SGP40* air quality sensors [see Fig. 1(e)].

3) Environmental Sensors: Two environmental *Bosch BME280* sensors [see Fig. 1(a)] were used to measure ambient air temperature, barometric pressure, and relative humidity. These provide a baseline indicator of the environmental conditions in which the fruit was ripening.

This sensors array was selected as they provided a cost-effective, readily available, and reliable data source, demonstrated by their use in previous works (see Section II). The cost of each sensor (VAT excl.) was as follows: Raspberry Pi Camera £11.81, MapIR Survey3N camera $400 USD, TRIAD spectrometer £54.21, Figaro TGS2602 £22.68, TGS2620 £24.50, SGP40 £10.77, and BME280 £12.99. The total cost of the sensors alone was therefore approximately £457.49.

(a) (b) (c) (d) (e)

Fig. 2. Example of the five-stage framework used in this work. (a) Stage 1. (b) Stage 2. (c) Stage 3. (d) Stage 4. (e) Stage 5.

TABLE II
Banana Ripeness Classification Descriptions

Stage:	Description of Banana's Appearance:
1 - Unripe	Whole banana is green.
2 - Ripe	Whole banana is yellow.
3 - Beyond ripe	Whole banana is yellow with some browning or brown spots that cover the minority of the visible surface.
4 - Over-ripe	Majority of the visible surface is brown or has brown spots.
5 - Gone off	Whole banana is black or brown.

B. Fruit Ripeness Categorization

A five-stage ripeness classification framework was chosen, instead of the seven-, four-, or three-stages identified in Section II, as five-stages allows sufficient precision to be captured without risking inaccurate classifications caused by an abundance of ripeness stages that are difficult to differentiate [7], [8], [9], [11], [40]. Table II describes the physical appearance of the banana for each ripeness stage under the experimental setup created for systematic data collection. Fig. 2 shows an example image of each ripeness stage taken from the test data.

C. Sensor Positions and Testing Chamber Layout

1) Rationale: The performance of the selected sensors is influenced by their placement around the banana being tested. The *MapIR* hyperspectral camera has a fixed, long focal length, requiring its subject to be some distance away to capture sharp images. The VOC sensors are ineffective when measuring low VOC concentrations, necessitating an enclosed chamber to limit dispersal during testing. The *Sparkfun spectrometer* requires controlled lighting conditions to accurately

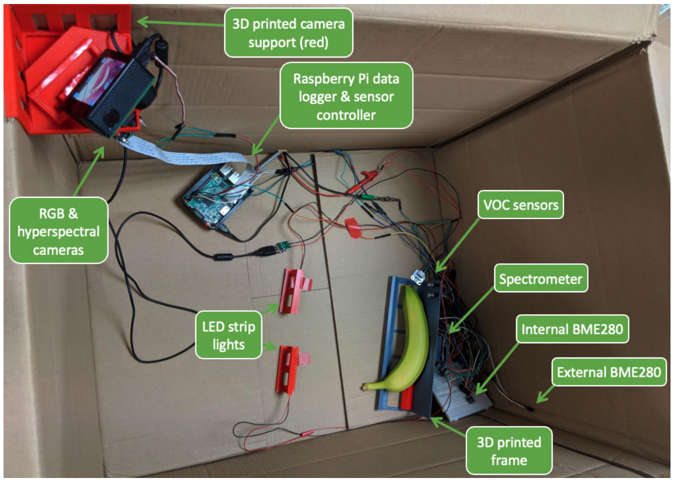

Fig. 3. Opaque cardboard test chamber for tests 3 and 4. Sensor and banana positions labeled. Cameras supported by red 3-D printed frame at top of chamber. Other sensors and banana diagonally opposite at bottom of chamber. LEDs mounted in red 3-D printed supports. ADC's and BME280's on white breadboard behind banana. Gray 3-D printed frame houses spectrometer to prevent optical interference.

and consistently measure banana skin reflectance, as external light or internal reflections can generate inconsistent readings.

2) Proposed Solution: A small polypropylene box, covered with a dark cloth was used for tests 1 and 2. However, its small size did not provide sufficient focal-distance for the cameras, resulting in blurry images. The *SGP40* air quality sensor was not installed and a low-quality analog to digital converter (ADC) was used to measure data from the two *Figaro* gas sensors, saturating early in the tests. There was no external *BME280* sensor, and the spectrometer was poorly supported and moved away from the banana throughout the tests. To solve this, a large, opaque cardboard box was sourced for tests 3 and 4. This allowed the two cameras to be placed diagonally opposite the target fruit, maximizing the available focal distance (see Fig. 3). The *SGP40* air quality, external *BME280*, and higher quality ADCs were used for the *Figaro* VOC sensors. light emitting diodes (LEDs) were used to illuminate the banana when the cameras were active. The *Sparkfun spectrometer* was mounted at the end of a 2-in 3-D printed shroud, perpendicular to the banana. Three gaseous sensors were placed close to the banana to ensure accurate measurement. The box was wrapped with film to prevent VOCs escaping from the chamber. One of the environmental *BME280* sensors was placed in the testing chamber and the other outside to account for external environmental fluctuations. As a result, the recorded data for tests 3 and 4 showed more variance and fewer plateaus than tests 1 and 2.

For all tests, the sensors were centrally controlled by a Raspberry Pi 3B Single Board Computer (SBC) via I2C. A Python script was written to collect data from all sensors, while ensuring the appropriate LED illumination (e.g., front flood illumination for RGB and hyperspectral image collection or white, IR or ultraviolet (UV) illumination for the spectrometer) was active. RGB images from the Pi Cam were stored locally on the Pi, while hyperspectral images were stored on the MapIR camera. All recorded data were logged locally on the Pi, in CSV format, while limited data

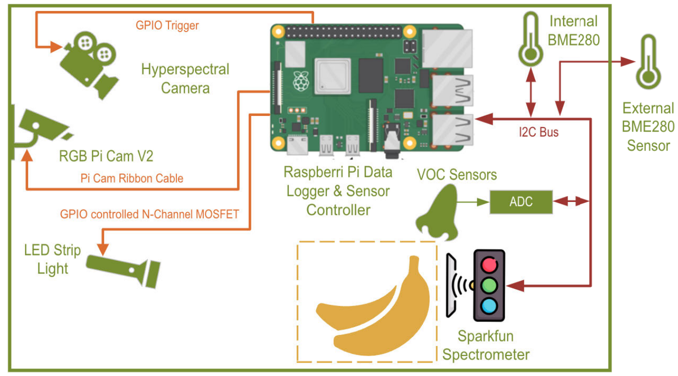

Fig. 4. Test chamber sensor schematic showing sensors (green), data buses (dark red), and sensor control connections (orange).

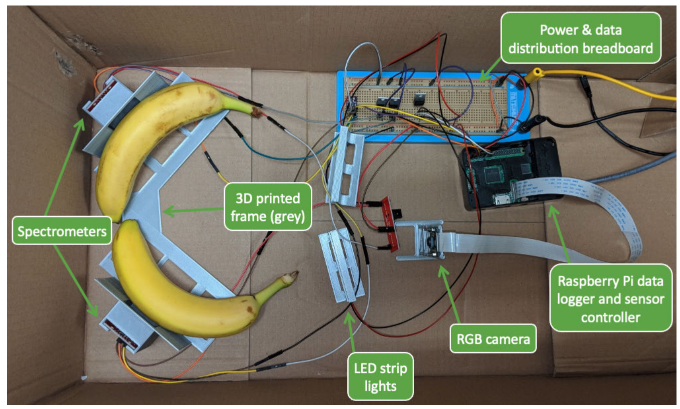

Fig. 5. Third cardboard chamber for testing two bananas simultaneously (Test 5). Sensor and banana positions labeled. Gray 3-D printed frame houses spectrometers to prevent optical interference. RGB camera and LEDs mounted on gray 3-D printed supports. Power, ADCs, BME280, and multiplexing on breadboard. Pi data logger in black case.

were uploaded to ThingSpeak to allow remote monitoring of the testing environment. A data sample from all sensors was measured and logged once every 15 min, starting on the hour. This was managed using `Crontab`. Fig. 4 provides a schematic of the sensor layout and connections within the test chamber.

3) Third Test Chamber: Preliminary data processing indicated that the spectral data from the *Sparkfun spectrometer* was the most useful in determining the banana's ripeness. Therefore, to collect more data rapidly, a third test chamber was assembled that would allow the spectral response of two bananas to be measured simultaneously. This test chamber included four sensors; two *Sparkfun spectrometers*, one BME280 environmental sensor (located internally), and one RGB camera. The test chamber shown in Fig. 3 was still operating, allowing a total of three bananas to be measured simultaneously. Fig. 5 shows the configuration of the third test chamber. A Raspberry Pi model 3B was used to log data and control the sensors and LEDs.

D. Environmental Variations and Outlier Data

The test chamber's external environment was poorly thermally insulated from the outdoors. This resulted in significant variations in internal chamber temperature, matching the

TABLE III
TEST DURATIONS AND DATA SAMPLE QUANTITIES

Test Number	1	2	3	4	5
Test Duration:	18 days 16:45 hrs	17 days 21:45 hrs	49 days 17 hrs	64 days 21:30 hrs	29 days 0:30 hrs
Number of Data Samples:	1796	1720	4777	6133	2516

weather of the season. This affected the speed of fruit maturation, and thus test length, as shown in Table III, which details the duration of each test and the number of data samples recorded. Comparisons between the internal and external environmental data indicated that the test chamber's humidity was stable, but otherwise matched the external environment's air pressure and temperature. On a number of days, the external and internal environment's temperature fluctuated significantly following the use of air conditioning. This caused significant fluctuations in measured VOC concentrations only.

E. Data Processing

1) Data Preparation: Prior to any analysis using the ML methods mentioned in Section III-F, the raw data were cleaned and preprocessed. Limits were enforced upon each sensor's data based on the measurement ranges specified in the sensor's datasheet. Data outside of these ranges was replaced with the mean of surrounding values. Scatter plots of data from each sensor were produced both before and after data preparation to ensure all anomalies were removed correctly, as shown in Fig. 9 (cleaned data only). Fruit ripeness stage numbers were generated for each data sample using one-hot encoding, based upon the methodology described in Section III-B.

2) Dataset Creation: As the data were collected in individual test groups, with a differing test chamber and assortment of sensors in each test, several datasets were created that grouped data based on the sensors used, the reliability of data collected from each sensor, and predetermined data-pairings of interest. Data from certain sensors during specific tests was suspected of being erroneous based on test chamber and/or hardware limitations noted at the time of measurement, and as such were excluded from the datasets (see Table IV). Details of the datasets, including their data sources, are shown in Table V.

The observed erroneous data were caused by poor quality ADCs introducing signal conditioning errors, insufficient current provision to some sensors and drifting sensor positions. These errors were observed and corrected for subsequent tests.

3) Removal of Class-Bias in Datasets: The variation in test lengths and number of data samples (Table III) created an unbalanced number of data samples for each ripeness stage. To avoid biasing the ML methods with unbalanced data, the number of data samples for each ripeness class were equalized using the `Synthetic Minority Oversampling Technique`. This generates synthetic samples for minority classes until their numbers are equal to that of the majority class. The balanced data were split into training (70%) and testing (30%) datasets using `Stratified Shuffle Split`, which ensured the class balance was maintained postsplit.

TABLE IV
PERTEST SENSOR UTILIZATION AND DATA QUALITY

Sensor:	Test Number: 1	2	3	4	5
BME280 - internal	Good	Good	Good	Good	Good
BME280 - external	N/A	N/A	Good	Good	N/A
VOC - TGS2620 & TGS2602	Poor	Poor	Good	Good	N/A
VOC - SGP40	N/A	N/A	Good	Good	N/A
SparkFun TRIAD Spectrometer	Poor	Good	Good	Poor	Good

TABLE V
INDIVIDUAL DATASET'S CONSTITUENT DATA

Dataset Name:	Sensor Data Included:	From Test Numbers:	Number of Samples:
TPHiS	BME280 - internal Spectrometry	2, 3, 4, 5	32665
Spect	Spectrometry	2, 3, 5	22695
34	All sensors	3, 4	26885
VOC	All VOC sensors	3, 4	26885

F. Data Analysis Methodology

1) Stage 1: Raw data were cleaned using the methodology described in Section III-E1. The cleaned data were plotted (e.g., Fig. 9) and statistical correlation matrices were calculated using the techniques described in Section III-F3 (e.g., Fig. 7). The scatter plots were examined for erroneous data and trends, which were amended as appropriate. The correlation matrices were examined for interesting inter-sensor relationships. Subsequently, the datasets were created in accordance with the methodology described in Section III-E2.

2) Stage 2: A series of ML algorithms, specified in Section III-F3, were trained to classify the ripeness of the banana using the measured data. Each algorithm was trained and tested using each dataset to determine which algorithms and datasets (and thus sensor(s)) were the most accurate for this classification task. The ANN was trained using the methodology shown in Section III-F3g. Four separate single-layer models were trained with 50 epochs each for each activation function and optimizer. Each model was tested against the test dataset and the accuracies recorded. This was repeated for a two-layer ANN.

3) ML Algorithms: In Section II and Table I, a number of ML algorithms were identified as relevant to the classification tasks in this article, such as ANNs, SVMs, KNNs, random forests (RFs), DTs, CNNs, DNNs, multilayer perceptrons (MLPs), gradient boosting classifiers (GBCs), Naïve Bayes, and principle component analysis (PCA) [10], [14], [24], [25], [26], [27]. The statistical methods pearson correlation coefficient (PCC) and Spearman's equivalent were used to identify basic correlations, as well as SVM, RF, k-Nearest Neighbors (KNN), GBC, and ANN for discrete classification. Fig. 6

illustrates the data-analysis pipeline used. A brief description of each implemented ML algorithm follows.

a) Pearson's correlation: PCC quantifies the linear relationship between two variables with a normalized output ranging from −1 to 1. The individual elements of a PCC matrix may be calculated using $r = (\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}))/(\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2)^{1/2}$ where x_i and y_i are the ith element and $\bar{x}$ and $\bar{y}$ are the means of data X and Y for n data points (i.e., the dataset size), where X is the data from one sensor and Y the data from another.

b) Spearman's correlation: Spearman's correlation measures the fit of a monotonic function to the relationship between two variables without assuming a linear relationship or normal distribution, unlike PCC. It is calculated as the PCC between the rank values of the variables using the formula $r_s = \rho_{R(X),R(Y)} = (\mathrm{cov}(R(X), R(Y)))/(\sigma_{R(X)}\sigma_{R(Y)})$, where ρ is the PCC calculated using rank variables, $\mathrm{cov}(R(X), R(Y))$ is the covariance of the rank variables, and $\sigma_{R(X)}$ and $\sigma_{R(Y)}$ are the standard deviations of the rank variables. X and Y represent data from different sensors.

c) SVM: SVMs are supervised learning methods for classification, regression, and outlier detection [41], excelling in high-dimensional spaces like that in this work. SVMs identify the optimal hyperplane that separates classes in the feature space with the greatest margin between nearest class points or support vectors. The hyperplane classification function is $f(x) = \mathrm{sign}(w \cdot x + b)$, where w is the weight vector perpendicular to the hyperplane, x the input features, and b the bias adjusting the hyperplane's origin distance. w and b are optimized during training. Kernel functions (e.g., `poly`, `RBF` and `sigmoid`) transform input data into a higher dimensional space for linear separation and class decisions, in which a positive result signifies class membership.

d) Random forest: RF is an ensemble learning method whereby a number of DTs are created during training, each resulting in a class output. During classification, the modal class is selected as the overall output, while regression selects the mean. Initially, multiple subsets of the original training dataset are sampled to train individual DTs. For prediction, test data are fed to all trees. If the decision of each tree is defined as $f_i(x)$ for the ith tree, where x are the input features, the classification model may be defined as $F(x) = \mathrm{mode}\{f_1(x), f_2(x), \ldots, f_N(x)\}$ where N is the number of trees. The regression model may be defined as $F(x) = (1/N)\sum_{i=1}^{N} f_i(x)$.

e) KNN: KNN does not require training, instead processing all data at the time of prediction. KNN calculates the distance between the test data and all training data and makes a class prediction based on the class of the nearest training data in the feature space, assuming that the nearest points are the same class. The Euclidean distance is most often used to calculate inter-data distance in the feature space, and is defined as $d(x, y) = ((x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2)^{1/2}$ where x and y are data points in an n-dimensional space.

f) GBC: GBC builds an ensemble of weak predictive models (e.g., DTs) to create a strong classifier. Initially, a model is fit to the data and the residuals calculated. Each subsequent model is trained on these residuals, improving the previous model's errors. The outputs of the individual models are summed to generate the final prediction, weighted by their learning rates, as described by the equation $F(x) = F_0(x) + \sum_{t=1}^{T} \gamma_t h_t(x)$, where F_0 is the initial model, γ_t the learning rate, and h_t the subsequent weak learners' contributions.

g) ANN: ANNs consist of neuron layers, each with activation function $y = f(\sum_{i=1}^{n} x_i w_i + b)$, where x_i, w_i, and b represent the input, weight, and bias of the ith neuron, respectively. Data are passed into the first layer and processed through each neuron's activation function, with outputs becoming subsequent neurons inputs. Weights and biases are adjusted during training via methods like backpropagation. Popular activation functions are *sigmoid*, *hyperbolic tangent* (`tanh`), and *rectified linear unit* (`ReLU`—preferred for CNNs), while *Adam* and *Stochastic Gradient Descent (SGD)* are optimizers.

IV. Experiments and Results

A. Overall Measurement Outcomes

1) VOC Measurements: As each test progressed, the concentration of measurable VOCs increased, as shown in Fig. 9(d). Fluctuations in measured VOC concentration coincided with fluctuations in temperature and humidity.

2) Spectral Measurements: Greater variation in spectral response was observed across all 18 measured wavelengths during the immature stages of fruit ripeness, as shown in Fig. 9(b). As the banana entered the later stages of ripeness, the spectral responses stabilized significantly, with no wavelength exceeding a value of 2.0 for approximately 24 h.

3) Environmental Measurements: The ambient temperature, pressure, and relative humidity recorded provide little information on their own, other than to indicate that the test chamber provides good humidity stabilization against the external environment. The internal and external air pressure are near-identical throughout testing, while the internal ambient temperature was marginally warmer than the external temperature, as shown in Fig. 9(c).

B. Basic Correlation Analysis

Pearson and Spearman correlation matrices were generated for all four datasets. The matrix for the "34" dataset is shown in Fig. 7. Particular measurement correlations were anticipated, such as strong positive correlations between internal and external environmental measurements. These were observed for all three parameters, with a correlation factor of 0.987, 0.999, and 0.683 for temperature, pressure, and humidity, respectively. A mild negative correlation between temperature and pressure and any VOC measurement was also observed, averaging −0.327, with the strongest correlation being −0.568 for the *SGP40* VOC sensor. Temperature and pressure showed a mild positive correlation with all spectral wavelengths, averaging to 0.278. There was a notable negative correlation between all VOC data (especially that recorded by both *Figaro* sensors) and the spectral data, averaging −0.475 (which rises to −0.548 when considering only the *Figaro* sensors). This correlation was most significant around the 510 nm (green) and neighboring wavelengths, at −0.886. A strong

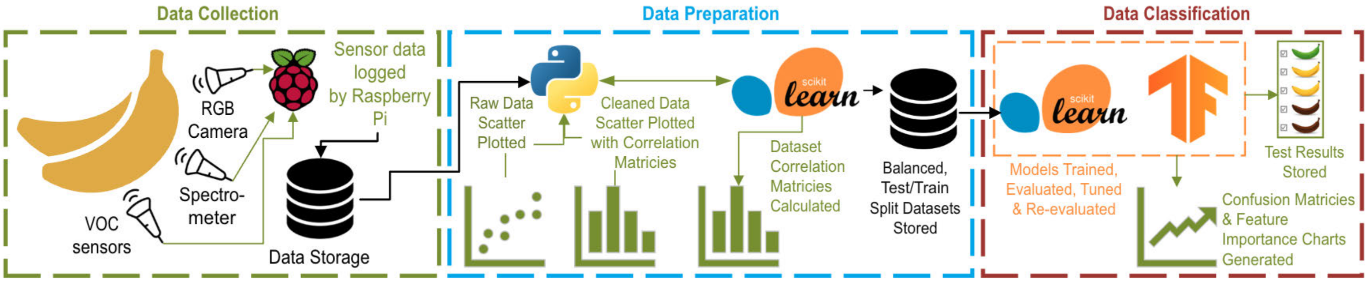

Fig. 6. Data analysis pipeline: green dashed box (left) depicts data collection process, blue dashed box (middle) depicts data preparation, and red dashed box (right) depicts ML training and testing.

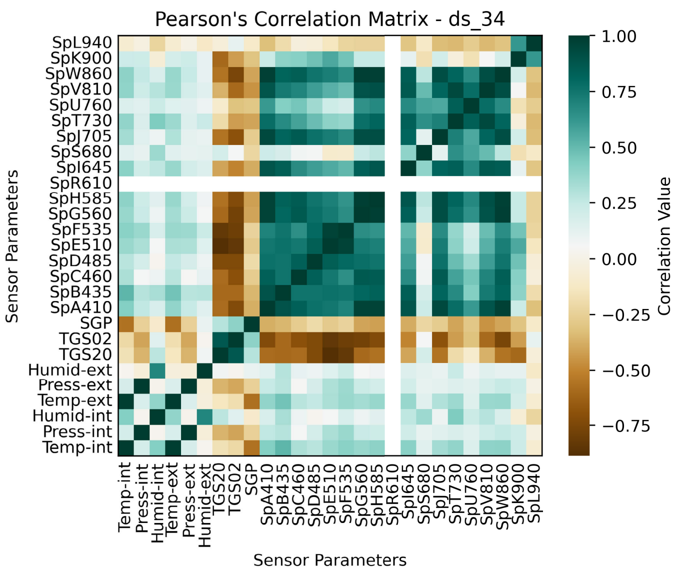

Fig. 7. “34” dataset Pearson’s correlation matrix.

positive correlation between most spectral wavelengths was also noted, with the exception of 680 nm (red), 900 nm, and 940 nm (both IR).

The Pearson correlation matrix for the “TPHiS” dataset (not printed) showed a negative correlation between temperature and all spectral wavelengths, averaging −0.235. The other environmental parameters showed no correlation. All spectral measurements were strongly positively correlated to each other (averaging 0.914). Otherwise, PCC and Spearman’s correlation matrices show similar correlations for the “TPHiS” dataset.

A fault with the Sparkfun spectrometer resulted in a single value being recorded for all 610 nm responses for tests 3 and 4, rendering the calculation of Pearson’s correlation impossible with data from the “34” dataset. As a result, only white (i.e. 0.00 correlation value) is shown for *SpR610* in Fig. 7.

C. Confusion Matrices

In confusion matrices showing the number of correct classifications during testing were generated for each ML method, against each dataset, both before and after hyperparameter optimization. The optimized results for each method were normalized to a percentage and averaged across each dataset to create a performance metric for each ML method (see Fig. 8). The SVM results show the RBF kernel as the most accurate. From Fig. 8, the average accuracies for each ML method range from 98.35% (KNN) to 98.72% (RF). The raw accuracy values and three-most important features for each ML algorithm tested against each dataset are shown in Table VI.

1) Analysis of Results: From Table VI, it is clear that the *polynomial* and *sigmoid* kernels for SVM achieve low accuracies regardless of the dataset they are trained and tested on. Model tuning was not carried out with either kernel (or a linear kernel) as the training failed to complete in a reasonable time-frame or produced negligible improvements to model accuracy. The RBF kernel achieved good accuracy across all datasets, ranging from 96.29% (“VOC”) to 99.89% (“TPHiS”). Where VOC data were available, the SVM relied upon this to make predictions, otherwise using 410 nm (violet), 535 nm (green), and 860 nm (NIR) spectral responses. Across all ML algorithms used, the highest accuracy achieved was 99.95% by RF and GBC with the “34” dataset, followed closely by KNN with 99.93% and the “TPHiS” dataset, with which RF and GBC also achieved 99.91%. Notably, the 535 nm (green) spectral response was a key predictor in both results achieved by RF and GBC.

Regarding the ANN, the *Adam* optimizer outperformed SGD in almost all cases. The best accuracy achieved by the single-layer ANN was 94.72% using the *sigmoid* activation function and *Adam* optimizer with the “Spect” dataset, averaging 94.00% across the other activation functions for this optimizer and dataset. Across all four datasets, the *Adam* optimizer was on average 28.25% more accurate than the SGD optimizer, regardless of activation function, with its own accuracy ranging from 66.85% (tanh) to 68.96% (relu). The best accuracy achieved by the two-layer ANN was 96.07% using *relu* and *sigmoid* activation functions on the first and second layers, respectively, with *Adam* optimizer and “Spect” dataset. Across all activation function configurations, the *Adam* optimizer and “Spect” dataset achieved the greatest accuracies, averaging 95.42%. Using the *relu* activation function in all the layers with the *Adam* optimizer yielded the greatest accuracy across all datasets at 78.89%, followed by *sigmoid* (first layer) *relu* (second layer) at 78.15%. When considering datasets alone, the “Spect” dataset yielded the most accurate results when training an ANN. When training SVM (RBF kernel only), RF, KNN, or GBC, the “TPHiS” and “34” datasets

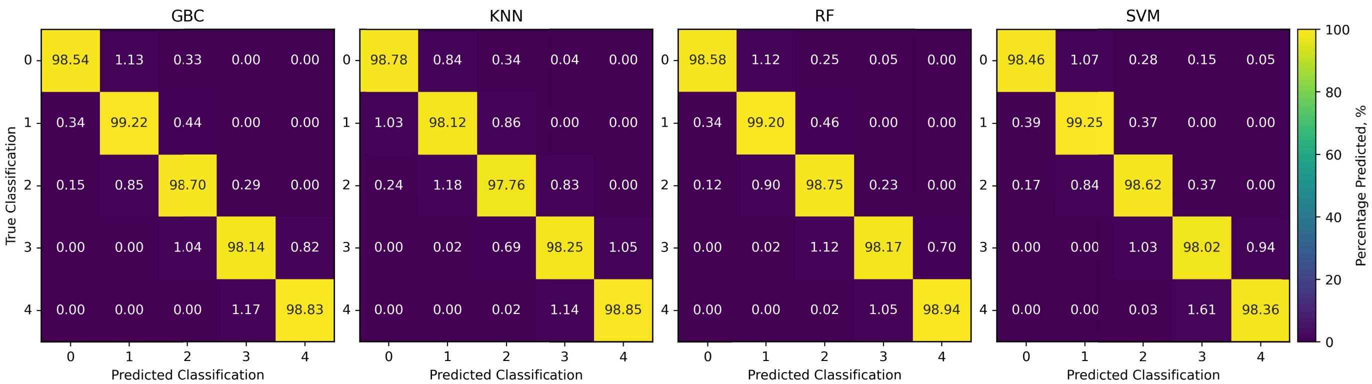

Fig. 8. Confusion matrices for each ML algorithm, averaged across each dataset.

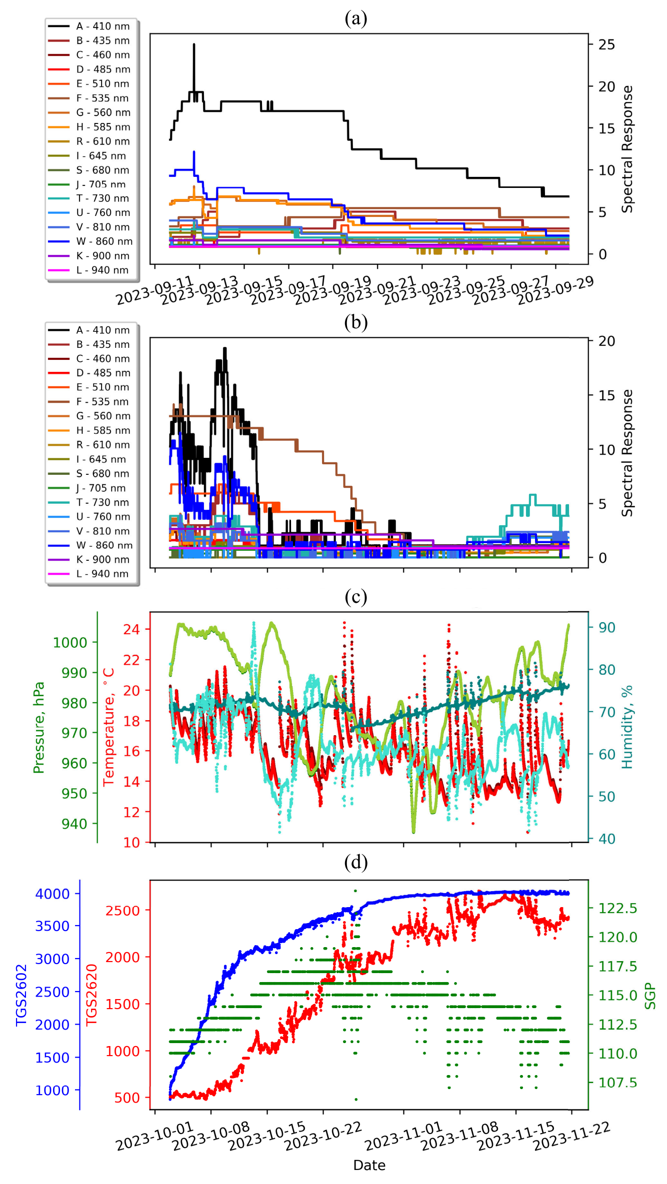

Fig. 9. Cleaned data plots. (a) Test 2 spectral response using first test chamber. (b) Test 3 spectral response using improved test chamber. (c) Test 3 internal and external environmental measurements. (d) Test 3 VOC measurements. All plots against date.

performed best, achieved 99.91% and 99.90%, respectively, on average.

TABLE VI
ML MODEL'S ACCURACY RESULTS AND THREE MOST IMPORTANT FEATURES FOR EACH DATASET AND ML ALGORITHM. SVM (POLYNOMIAL AND SIGMOID KERNEL) WERE UNOPTIMIZED. ALL OTHER RESULTS GENERATED USING OPTIMIZED MODELS

	Dataset Name:							
	TPHiS		**Spect**		**34**		**VOC**	
	Acc.	*Best 3 Features*	*Acc.*	*Best 3 Features*	*Acc.*	*Best 3 Features*	*Acc.*	*Best 3 Features*
SVM - Poly.	46.64%	A - 410 nm Pressure(int) W - 860 nm	75.55%	A - 410 nm B - 435 nm G - 560 nm	82.09%	TGS2620 TGS2602 Humidity(ext)	81.14%	TGS2620 TGS2602
SVM - RBF	99.89%	A - 410 nm F - 535 nm W - 860 nm	98.13%	A - 410 nm W - 860 nm F - 535 nm	99.85%	TGS2620 TGS2602 Humidity(ext)	96.29%	TGS2620 TGS2602 SGP40
SVM - Sigmoid	28.52%	A - 410 nm Humidity(int) F - 535 nm	52.15%	W - 860 nm F - 535 nm B - 435 nm	19.99%	TGS2620 Pressure(int) Pressure(ext)	19.99%	SGP40 TGS2620 TGS2602
Random Forest	99.91%	F - 535 nm E - 510 nm Pressure(int)	98.19%	W - 860 nm B - 435 nm F - 535 nm	99.95%	TGS2602 TGS2620 F - 535 nm	96.90%	TGS2602 TGS2620 SGP40
KNN	99.93%	N/A	97.87%	N/A	99.85%	N/A	95.77%	N/A
Gradient Boosting Classifier	99.91%	F - 535 nm Pressure(int) G - 560	98.28%	F - 535 nm B - 435 nm L - 940 nm	99.95%	F - 535 nm TGS2602 TGS2620	96.62%	TGS2620 TGS2602 SGP40
	Acc.	*Activation & Optimiser*	*Acc.*	*Activation & Optimiser*	*Acc.*	*Activation & Optimiser*	*Acc.*	*Activation & Optimiser*
ANN (1 layer)	77.13%	relu adam	94.73%	sigmoid adam	87.71%	tanh adam	52.07%	sigmoid adam
ANN (2 layers)	85.17%	sig-relu adam	96.08%	relu-sig adam	94.95%	sig-sig adam	66.43%	relu-sig adam

V. DISCUSSION

The result noted in Section IV-A1 match expectations that the banana would continue to produce VOCs throughout its ripening process (thus increasing VOC concentration in the test-chamber), the concentration of which is effected by ambient temperature and humidity, as these environmental factors are already known to effect ripening speed. In an enclosed chamber that remains sealed for the duration of testing, an approximately linear correlation with fruit ripeness can be established. This is not feasible outside of the laboratory where fruit-generated VOCs cannot be contained. Another aspect for further investigation is the use of other materials for the test chamber. While cardboard was used in this work, other materials, such as metals or plastics, may be trialed to note their effect on the absorption of the generated VOCs and thus their effect on the recorded accuracies.

When training ML models with data from all sensors (i.e. dataset "34"), the VOC data were the most influential in predicting the banana's current ripeness stage. Therefore, datasets that exclude VOC data may provide a more interesting insight into ripeness classification, as sensor data less tangibly linked to banana ripeness must be used in the prediction. Similarly, this could be extended to suggest that the "VOC" dataset, containing only VOC data, would provide the most accurate ripeness stage predictions. However, the results in Table VI indicate generally the opposite, likely because this dataset contains data from only three sensors. When examining the ML test results from datasets without VOC data (i.e., "TPHiS" and "Spect") wavelengths F (535 nm—green), A (410 nm—violet), W (860 nm—IR), and B (435 nm—violet) are most useful in classification predictions across all ML algorithms, suggesting that the proportion of green pigment within the banana's flesh is a key predictor for banana ripeness, as well as responses beyond the visible spectrum.

A. Algorithm Specific Results

The RF and GBC methods were identified as the most accurate at predicting fruit ripeness at 99.95% accuracy each. This was achieved using the "34" dataset, which contains the most features (27 features). The next most accurate classifications were achieved by the same algorithms in addition to KNN at 99.91% and 99.93% accuracy, respectively, using the "TPHiS" dataset, which contains the second greatest number of features (21 features). It follows that classification accuracy is linked to the number of features within the dataset. However, the performance improvement over a dataset such as "VOC," which contains only three features, is marginal as accuracies as high as 96.90% were achieved using that dataset with the RF algorithm.

The same correlation is not strictly true when considering the test results from the trained ANN, which exhibits the highest accuracy with the "Spect" dataset [94.73% (single layer) and 96.08% (two-layer)], and contains only the third greatest number of features 18. The more feature-rich "34" dataset achieves the second best accuracies across the datasets, while, as anticipated, the "VOC" dataset (three features) achieves a poor classification accuracy of 66.43%. Therefore, it is evident that the RF, GBC, and KNN ML algorithms provide a superior classification accuracy compared to the trained ANNs.

B. Comparisons With the Literature and Future Work

The RF model presented in this work outperforms those specified in [14], [24], and [27] by 2.20%–9.95%, while the GBC model outperforms that described in [26] by 8.70%. The KNN model outperforms those in [14], [24], and [26] by 3.33%–17.43%. Our ANN models outperform the ANNs described in [10], [22], [27], and [29] by 0.58%–12.08%, despite [27], [29] utilizing a DNN architecture. However, the ANN presented here is outperformed by the work in [14] by 1.67%.

The work presented in this article leaves scope for improvements, such as the collection of a larger dataset, the use of a more environmentally isolated test chamber, the incorporation of alternative VOC sensors, capturing and analysis of a wider spectral range of data, analysis of the hyperspectral images collected, analysis using additional ML algorithms, real-time "online" ripeness classification, and the creation of a compact field-ready hardware system for use outside of the laboratory.

VI. Conclusion

The most accurate ML method for predicting banana ripeness was RF and GBC equally with 99.95% accuracy, using the "34" dataset that contains data from all employed sensors, totaling 27 features. The next most accurate result was achieved with the "TPHiS" dataset, containing 21 features, using KNN at 99.93%, followed closely by RF and GBC at 99.91% with the same dataset. This work demonstrated that low-cost, readily available sensors can be used to accurately predict the banana ripeness without the destruction of the banana under test. The sensors and ML methods used may be easily deployed in a handheld, portable device that could be used by fruit growers to determine the banana ripeness in real time, prior to harvesting. This near-instant access to accurate ripeness information may help growers to pinpoint optimal harvesting time, ensuring fruits arrive at market in their optimal condition, reducing the number of fruits that must be discarded before sale and reducing food waste.

Data Availability

Data created in this research work is openly available from the University of Bath Research Data Archive at https://doi.org/10.15125/BATH-01459.

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